

Statistics: Mean & Median — 30 MCQs

Digital SAT Style | Easy / Medium / Hard

Q1. DIFFICULTY: EASY

The scores of 7 students on a quiz are 12, 15, 18, 20, 22, 22, and 31. What is the median score?

- A) 18
 - B) 20
 - C) 22
 - D) 24
-

Q2. DIFFICULTY: EASY

Data set: 4, 9, 9, 13, 15

What is the mean of this data set?

- A) 9
 - B) 10
 - C) 11
 - D) 13
-

Q3. DIFFICULTY: EASY

Five friends recorded their hours of sleep: 6, 7, 7, 8, and 12. Which statement is true?

- A) Mean > Median
 - B) Mean = Median
 - C) Mean < Median
 - D) Median = 7.5
-

Q4. DIFFICULTY: EASY

The heights (in inches) of 6 students are 60, 62, 64, 66, 68, and 70. What is the median height?

- A) 63
 - B) 64
 - C) 65
 - D) 66
-

Q5. DIFFICULTY: EASY

A teacher records test scores: 72, 80, 84, 88, 96. What is the median?

GRID-IN

[GRID-IN — enter your answer in the box]

Q6. DIFFICULTY: EASY

The following data set has a mean of 10:

6, 8, 10, x , 14

What is the value of x ?

- A) 10
 - B) 12
 - C) 14
 - D) 16
-

Q7. DIFFICULTY: EASY

Data set A: 3, 7, 7, 9, 14

Data set B: 3, 7, 7, 9, 14, 25

Which statement best compares the medians?

- A) Median of A is greater than median of B
 - B) Median of A is less than median of B
 - C) The medians are equal
 - D) Not enough information to compare
-

Q8. DIFFICULTY: EASY

A set of 4 numbers has a mean of 15. Three of the numbers are 12, 14, and 18. What is the fourth number?

- A) 14
 - B) 15
 - C) 16
 - D) 17
-

Q9. DIFFICULTY: EASY

What is the median of: 5, 5, 6, 8, 11, 13?

- A) 6
 - B) 7
 - C) 8
 - D) 9
-

Q10. DIFFICULTY: EASY

A data set has exactly 5 values. When arranged from least to greatest, what position is the median?

- A) 1st

- B) 2nd
 - C) 3rd
 - D) 4th
-

Q11. DIFFICULTY: MEDIUM

The mean of 8 numbers is 24. When a 9th number is added, the new mean becomes 22. What is the 9th number?

- A) 4
 - B) 6
 - C) 8
 - D) 10
-

Q12. DIFFICULTY: MEDIUM

Data set: 3, 7, 9, 12, 14, 18

If the value 18 is removed, which of the following is true?

- I. The mean decreases.
- II. The median decreases.

- A) I only
 - B) II only
 - C) Both I and II
 - D) Neither I nor II
-

Q13. DIFFICULTY: MEDIUM

A data set of 6 values has a median of 14. The values in order are: 8, 11, a, b, 19, 24. What is the value of $a + b$?

GRID-IN

[GRID-IN — enter your answer in the box]

Q14. DIFFICULTY: MEDIUM

The table below shows the number of books read by students:

Books read: 1, 2, 3, 4, 5

Frequency: 4, 6, 8, 5, 2

What is the mean number of books read?

- A) 2.4
- B) 2.8
- C) 3.0
- D) 3.2

Q15. DIFFICULTY: MEDIUM

A set of 7 values has a mean of 30 and a median of 28. A new value of 28 is added. Which must be true?

- A) The new mean equals 30.
 - B) The new mean is less than 30.
 - C) The new median equals 28.
 - D) The new median is greater than 28.
-

Q16. DIFFICULTY: MEDIUM

Package weights (in pounds): 14, 22, 35, 12, 8, 31, 22, 9
What is the median weight?

GRID-IN

[GRID-IN — enter your answer in the box]

Q17. DIFFICULTY: MEDIUM

A company tracks complaints per day for 10 days: 2, 3, 3, 5, 5, 7, 8, 9, 10, 12. The company removes the highest value (12). Which is true?

- A) Both mean and median decrease.
 - B) Mean decreases but median stays the same.
 - C) Median decreases but mean stays the same.
 - D) Both stay the same.
-

Q18. DIFFICULTY: MEDIUM

If the mean of {6, 9, 12, k, 18} equals the median, what is the value of k?

- A) 10
 - B) 12
 - C) 15
 - D) 18
-

Q19. DIFFICULTY: MEDIUM

Commute times (minutes) of 5 employees: 20, 35, 35, 40, 70. Which is true?

- I. The mean is greater than the median.
 - II. Removing 70 would decrease the median.
- A) I only
 - B) II only
 - C) Both I and II
 - D) Neither I nor II
-

Q20. DIFFICULTY: MEDIUM

Seven consecutive even integers have a mean of 20. What is the median?

GRID-IN

[GRID-IN — enter your answer in the box]

Q21. DIFFICULTY: MEDIUM

Data set P: 10, 12, 14, 16, 18 (mean=14, median=14).

A constant of 4 is added to every number to create Data set Q. Which is true about Q?

- A) Mean=14, Median=14
 - B) Mean=18, Median=14
 - C) Mean=18, Median=18
 - D) Mean=14, Median=18
-

Q22. DIFFICULTY: MEDIUM

A bakery sells 5 types of pastries. The mean price is \$4.20. Four prices are \$3.00, \$4.00, \$4.50, and \$5.50. What is the price of the fifth pastry?

- A) \$4.00
 - B) \$4.20
 - C) \$4.50
 - D) \$5.00
-

Q23. DIFFICULTY: HARD

The mean of a data set of n values is 40. When 3 values each equal to 10 are added, the new mean is 34. What is the value of n ?

- A) 3
 - B) 6
 - C) 9
 - D) 12
-

Q24. DIFFICULTY: HARD

Data set: 5, 8, 11, x , 17, 20 (in increasing order, x is a positive integer). If the median equals the mean, what is the value of x ?

GRID-IN

[GRID-IN — enter your answer in the box]

Q25. DIFFICULTY: HARD

A data set of 10 values has a mean of 50 and a median of 48. An outlier with value 120 is removed. Which is most likely true?

- A) Both mean and median decrease significantly.
 - B) Mean decreases significantly; median changes little or not at all.
 - C) Mean stays the same; median decreases significantly.
 - D) Both stay the same.
-

Q26. DIFFICULTY: HARD

Fifteen students took an exam. The mean score of all 15 is 72. The mean of the 10 who passed is 80. What is the mean score of the 5 who failed?

- A) 48
 - B) 50
 - C) 56
 - D) 64
-

Q27. DIFFICULTY: HARD

A data set has 12 values with a mean of 25 and a median of 22. A new value of 22 is added as the 13th value. Which must be true?

- I. The mean decreases.
- II. The median remains 22.
- III. The mean remains 25.

- A) I only
 - B) II only
 - C) I and II
 - D) III only
-

Q28. DIFFICULTY: HARD

A class of 20 students has a mean score of 80 on a test. The teacher adds 2 points to every score, then adds a new student who scored 82. What is the new mean for the 21 students?

GRID-IN

[GRID-IN — enter your answer in the box]

Q29. DIFFICULTY: HARD

A data set with an even number of values has its two middle values equal to 16 and 24 when sorted. A new value of 20 is added. Which must be true about the new median?

- A) The new median is 20.
 - B) The new median is less than 20.
 - C) The new median is greater than 20.
 - D) It could be 20, less than 20, or greater than 20.
-

Q30. DIFFICULTY: HARD

In a data set of 9 values, the mean is 36 and the median is 30. The largest value is 90. If the largest value is replaced with 60, which is true?

- A) Both mean and median decrease.
 - B) Mean decreases; median stays the same.
 - C) Median decreases; mean stays the same.
 - D) Both stay the same.
-